

# Macroeconomic Theory and Policy: Lecture 6 (Ch. 5)

# What are You Going to Learn (Ch. 5)

- Solow Growth Model
- How capital accumulates over time
- How diminishing marginal product of capital explains differences in growth rates across countries
- The principle of transition dynamics
- The limitations of capital accumulation, although a significant part of economic growth is still left unexplained

# Solow Growth Model

- The Solow growth model builds on the production model
  - ▶ It provides us with a starting point to discuss why growth differs across countries
  - ▶ The model has been extended in a number of important directions
  - ▶ Foundation for modern economic growth theory (and a lot of empirical work)

# Solow Growth Model

- There are a few key differences:
  - ▶ Capital is endogenized: converted from an exogenous to an endogenous variable
  - ▶ Augments the production model with capital accumulation
    - ★ Agents in the model can accumulate tools, machines, computers, and buildings over time
  - ▶ The accumulation of capital is a possible engine of long-run economic growth
  - ▶ In other words, some countries are richer than others because they invest more in accumulating capital

# Setting Up the Model: Production Function

- Add an equation for the accumulation of capital over time to the production model
- The Cobb-Douglas production function:
  - ▶ Constant returns to scale
  - ▶ Increasing in both capital and labor
- Variables are time subscripted ( $t$ )

$$Y_t = F(K_t, L_t) = \bar{A}K_t^\alpha L_t^{1-\alpha}$$

The  $Y_t$  is now the output in period  $t$ ,  $K_t$  is the capital in period  $t$  and  $L_t$  is the labour used in period  $t$

# Setting Up the Model: Resource Constraint

- Output can be used for consumption or investment

$$Y_t = C_t + I_t$$

$C_t$  is the consumption in period  $t$ ,  $I_t$  is the investment in period  $t$

- Assumes no imports or exports (i.e., closed economy)

# Setting Up the Model: Capital Accumulation

- Goods invested for the future determine the accumulation of capital
- Capital accumulation equation:

$$K_{t+1} = K_t + I_t - \delta K_t$$

$K_{t+1}$  is the next year's capital

$K_t$  is the this year's capital

$I_t$  is this year's investment

$\delta K_t$  is the depreciation of capital

$\delta$  is the depreciation rate which usually assume to be  $0 < \delta < 1$

- We will use  $\bar{d}$  and  $\delta$  interchangeably. Typically, the estimate is around 7 to 10% annually

# Setting Up the Model: Capital Accumulation

- Change in capital stock defined as

$$\Delta K_{t+1} = K_{t+1} - K_t$$

then,

$$\Delta K_{t+1} = I_t - \delta K_t$$

Future capital depends on investment today

- Today's capital is a result of investments in the past
  - ▶ This applies for all time periods in the model other than the first
  - ▶ So, we assume the economy is endowed with some initial level of capital at  $t = 0$

# Setting Up the Model: Labour

- For simplicity, labor supply is assumed to be inelastic and is given endogenously

$$L = \bar{L}$$

- A simple extension would be allowing for population growth, which we will cover later

# Setting Up the Model: Investment

- The economy consumes a fraction of output and invests the rest

$$I_t = \bar{s} Y_t$$

where  $\bar{s}$  is the fraction of output invested ( $0 < \bar{s} < 1$ )

- This implies

$$C_t = (1 - \bar{s}) Y_t$$

Consumption is the share of output not invested

- We will go through a model of investment and consumption later in the term

# Five Equations and Five Unknowns

- Endogenous variables:  $Y_t, K_t, L_t, I_t, C_t$
- Parameters:  $\bar{A}, \bar{s}, \delta, \bar{L}, \bar{K}_0$
- Five equations:

$$Y_t = \bar{A}K_t^\alpha L_t^{1-\alpha}$$

$$\Delta K_{t+1} = I_t - \delta K_t$$

$$L_t = \bar{L}$$

$$Y_t = C_t + I_t$$

$$I_t = \bar{s}Y_t$$

- This is a dynamic model in which each equation holds at each point in time

# Some more Differences between Solow and Production Model

- In the Solow model, we have added dynamics in the form of capital accumulation
- The Solow model leaves out the markets for capital and labor and the corresponding prices—the rental price of capital ( $r$ ) and the wage rate ( $w$ )
  - ▶ This is a simplification that makes the model easier to solve
- The consumption equation is part of the model, but it would be redundant to include

# Variables: Stock and Flow

- Stock: A quantity that survives from period to period
  - ▶ Capital in Solow model is a stock variables
    - ★ e.g., Tractor, house, factory
  - ▶ Think of it as the saving you have in the saving amount
- Flow: A quantity that lasts a single period
  - ▶ Output, investment consumption are a flow variables
  - ▶ Think of it as the amount you just deposit to the saving account
- The change in a stock is a flow
  - ▶ Change in the capital stock equals the flow of investment

# Prices and the Real Interest Rate

- We can add the wage and rental rate to our model, we would have
  - ▶ Two more endogenous variables and two more equations

$$w = MPL \text{ and } r = MPK$$

- ▶ Omitting them does not change the basic behaviour of the model

# Saving in Solow Model

- National income identity also hold in the Solow model
- The saving is just the difference between income and consumption

$$\text{Saving} = Y_t - C_t = I_t = \text{Investment}$$

- Each unit of saving gets used as a unit of investment

$$\text{Saving} = \text{investment} = \text{supply of capital}$$

# Solving the Solow Model

- The model needs to be solved at every point in time, which cannot be done algebraically
  - ▶ There are infinite amount of  $t$
- Two ways to make progress:
  - ▶ Show a graphical solution
  - ▶ Solve the model in the long run
- Begin by combining equations algebraically

# Solving the Solow Model

- Convert  $Y$ ,  $K$  and  $I$  into per capita term

$$y_t = Y_t/L_t, k_t = K_t/L_t \text{ and } i_t = I_t/L_t$$

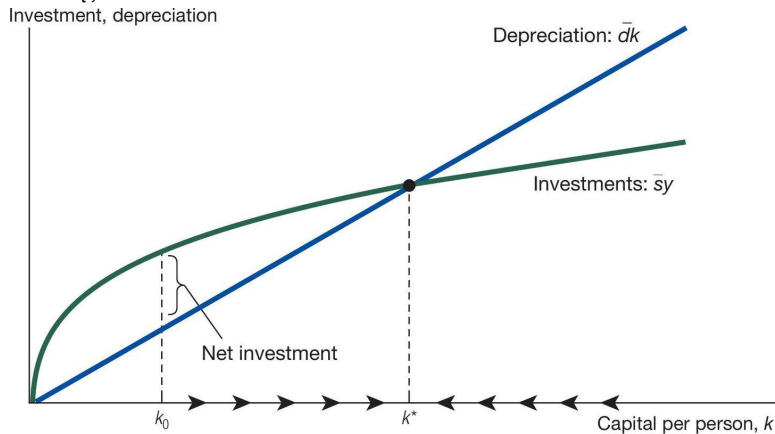
- ▶ Since we assume  $L = \bar{L}$ , we can solve the model at level as well (see extra question)
- Combine the investment allocation ( $i_t = \bar{s}y_t$ ) and capital accumulation equation ( $\Delta k_{t+1} = i_t - \delta k_t$ )

$$\Delta k_{t+1} = \bar{s}y_t - \delta k_t$$

Change in Capital per capita = Net Investment per capita

# The Solow Diagram

- Investment is  $i_t = \bar{s}y_t = \bar{s}\bar{A}k_t^\alpha$ 
  - ▶ with the usual assumption that  $0 < \alpha < 1$
  - ▶ increasing with diminishing return
- Depreciation is  $\delta K_t$ , which is linear



# Using the Solow Diagram

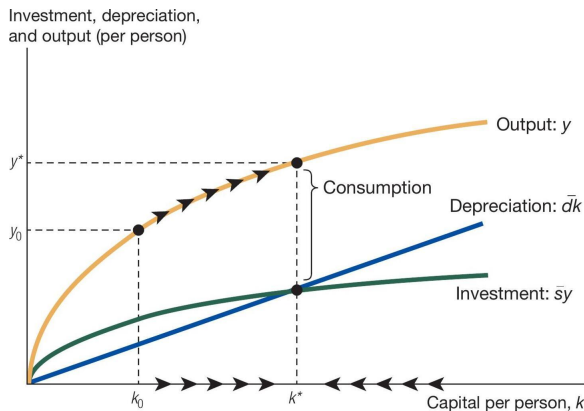
- If the amount of investment  $>$  depreciation,  $sy_t > \delta k_t$ 
  - ▶ Capital stock will increase until the amount of investment being undertaken is equal to the amount of capital that wears out through depreciation  $sy_t = \delta k_t$
  - ▶ This is called the steady state, capital stock will stay at this value of capital forever and the change in capital is equal to 0
- If depreciation is greater than investment,  $sy_t < \delta k_t$ , the economy converges to the same steady state as above
  - ▶ Capital stock will increase until the amount of investment being undertaken is equal to the amount of capital that wears out through depreciation

# Model Dynamics

- When not in the steady state, the economy exhibits a change in capital toward the steady state
- As  $k$  moves to its steady state, output will also move to its steady state
- At the steady state, all endogenous variables are steady
- *Transition dynamics* take the economy from any possible initial level of capital to the steady state
- No matter what level of initial capital we choose, if we wait long enough, the economy will converge to the steady state,  $k^*$

# The Solow Diagram with Output

- We can also plot the production function to see how output evolves over time
- As transition dynamics take the economy from  $k_0$  to  $k^*$  output rises over time from  $y_0$  to  $y^*$
- Consumption can also be plotted, it is the difference between output and investment



# Solving for the Steady State

- We can solve mathematically for the steady-state level of capital
  - ▶ Recall that steady state is a long-run equilibrium where, if the model's variables equal a certain set of values, they will never again change from these values
- In the steady state, investment equals depreciation (i.e.,  $\Delta K = 0$ )

$$\Delta k = 0 = \bar{s}y^* - \delta k^* \implies \bar{s}y^* = \delta k^*$$

$$y^* = \frac{\delta k}{\bar{s}}$$

# Solving for the Steady State

- Substitute into the production function and solve for  $K^*$

$$k^{*1-\alpha} = \frac{\bar{s}\bar{A}}{\delta} \implies k^* = \left( \frac{\bar{s}\bar{A}}{\delta} \right)^{1/(1-\alpha)}$$

- The steady state level of capital is positively related to the
  - ▶ investment rate ( $\bar{s}$ )
  - ▶ size of the workforce ( $\bar{L}$ )
  - ▶ productivity of the economy ( $\bar{A}$ )
- Negatively correlated with the depreciation rate ( $\delta$ )

# Solving for the Steady State

- To get  $y^*$ , plug  $k^*$  into the production function,  $y^* = \bar{A}k^{*\alpha}$

- We have

$$y^* = \bar{A} \left( \left( \frac{\bar{s}\bar{A}}{\delta} \right)^{1/(1-\alpha)} \right)^{\alpha} \implies y^* = \bar{A} \left( \frac{\bar{s}\bar{A}}{\delta} \right)^{\alpha/(1-\alpha)}$$

$$y^* = A^{\frac{1}{1-\alpha}} \left( \frac{\bar{s}}{\delta} \right)^{\frac{\alpha}{1-\alpha}}$$

- Higher steady state production caused by higher productivity and investment rate
- Lower steady state production caused by faster depreciation

# Solving for the Steady State

$$y^* = A^{\frac{1}{1-\alpha}} \left( \frac{\bar{s}}{\delta} \right)^{\frac{\alpha}{1-\alpha}}$$

- Note the exponent on productivity is different here than in the production model
  - ▶ With  $\alpha = 1/3$ ,  $\frac{1}{1-\alpha} > 1$
- Higher productivity has additional effects in the Solow model by leading the economy to accumulate more capital
  - ▶ In other word, the level of  $K$  depend on productivity
  - ▶ More output means more investment, which leads to higher levels of capital

# Understanding the Steady State

- Steady state eventually reaches when we have
  - ▶ a constant level of  $K^* = K_t = K_{t+1} = \dots$
  - ▶ a constant level of  $Y^* = Y_t = Y_{t+1} = \dots$
  - ▶ And all other variables,  $y, i, k, \dots$

# Understanding the Steady State

- The economy attains a steady state because:
  - ▶ Investment exhibits diminishing returns
  - ▶ The growth rate of production and investment diminishes as the capital stock grows
- Additionally, a constant fraction of the capital stock depreciates every period, with this depreciation rate remaining constant even as the capital accumulates
  - ▶ Depreciation is not subject to diminishing returns as capital increases
- Consequently, net investment eventually becomes zero, marking the point at which the economy enters a steady state.

# Economic Growth in the Solow Model

- Important result: there is still no long-run growth in the Solow model
- In the steady state, growth stop, and variables, total output (Y), capital stock(K), output per Capita (y), per capita consumption (c) and so forth remain constant
- The economy grows (or shrink) from  $k_0$  to  $k^*$ , but eventually growth stops as the capital stock and production converge to constant levels

# Economic Growth in the Solow Model

- Economies empirically exhibit continuous growth
- The model shows a limitation in explaining this growth
- According to the model, capital accumulation isn't the source of long-term economic growth
  - ▶ Savings and investment matter for growth in the medium run, but
  - ▶ In the long run, diminishing returns to capital accumulation cause the return on these investments to fall
  - ▶ However, they don't support long-term growth due to diminishing returns
- A possible explanation for this could be that we have not reached the steady state... but most economist would agree that there is something else that is driving the growth

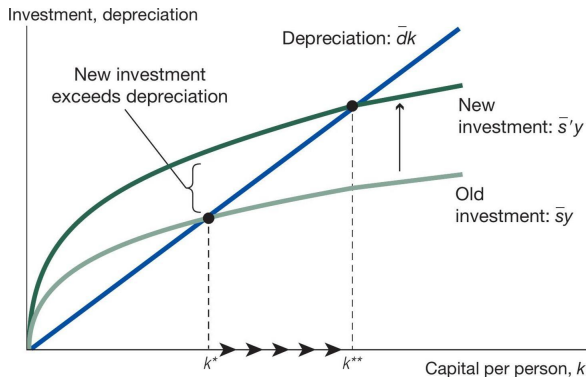
# Some Economic Experiments

- The Solow model
  - ▶ does not explain long-run economic growth
  - ▶ does help explain some differences across countries
- Economists can experiment with the model by changing parameter values

# An Increase in the Investment Rate

- Suppose the investment rate increases permanently for exogenous reasons ( $s' > s$ )

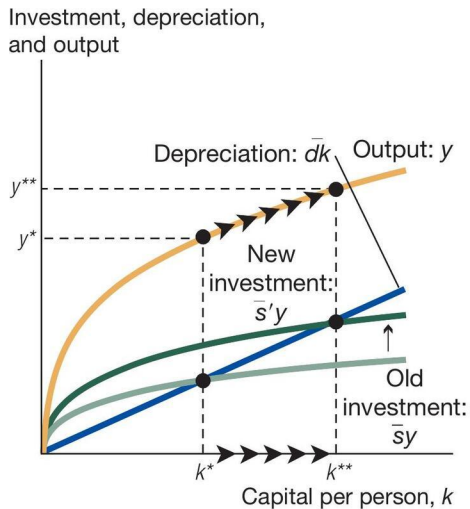
- ▶ The investment curve rotates upward ( $\bar{s}'\bar{A}k_t^\alpha > \bar{s}\bar{A}k_t^\alpha$ )
- ▶ The depreciation curve remains unchanged ( $\delta j_t$ )
- ▶ The capital stock increases by transition dynamics to reach the new steady state
  - ★ This happens because investment exceeds depreciation ( $sy_t > \delta k_t$ )
- ▶ The new steady state is located to the right



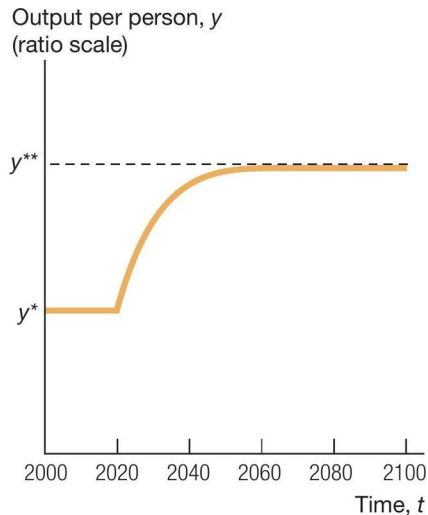
# An Increase in the Investment Rate

- What happens to output in response to this increase in the investment rate?
  - ▶ The rise in investment leads capital to accumulate over time
  - ▶ This higher capital causes output to rise as well
  - ▶ Output increases from its initial steady state level  $y^*$  to the new steady state  $y^{**}$
  - ▶ The increase in the investment rate causes the economy to grow over time, until a new steady state is reached
  - ▶ In the long run, both steady state capital and steady state production are higher
  - ▶ Because labour is constant, this also means the level of output per person is permanently higher as well

# The Behaviour of Output after an Increase in $\bar{s}$



(a) The Solow diagram with output.



(b) Output per person over time.

# Next Lecture

- Next lecture will finish Ch.5